MATHS · CLASS 6–7

Fractions

Notes · what they are, equal fractions, comparing

Name: _____

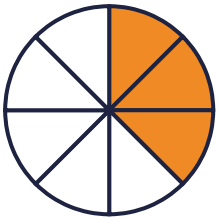
Date: _____

1. What is a fraction?

A fraction tells us **how many equal parts** we have out of the whole.

$\frac{3}{8}$ → 3 parts shaded, out of 8 equal parts

Numerator (top) = parts we take. **Denominator** (bottom) = equal parts in one whole.

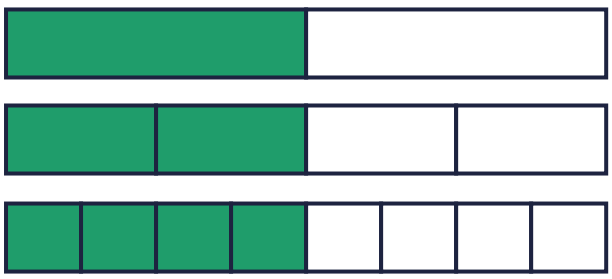




Remember The parts must be **equal**. A pizza cut into 4 unequal pieces is not quarters!

2. Equal (equivalent) fractions

Different fractions can show the **same amount**:



$\frac{1}{2}$

$\frac{2}{4}$

$\frac{4}{8}$

To make an equal fraction, **multiply (or divide) the top and the bottom by the same number**:

$\frac{1}{2} = \frac{1\times2}{2\times2} = \frac{2}{4}$ $\quad \frac{18}{24} = \frac{18\div6}{24\div6} = \frac{3}{4}$ (simplest form)

3. Comparing fractions

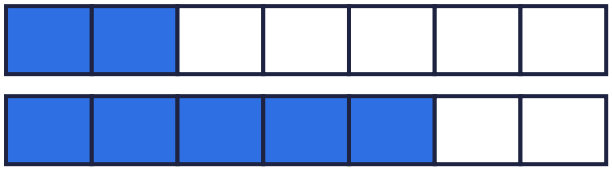
- **Same bottom:** bigger top wins. $\frac{5}{7} > \frac{3}{7}$
- **Same top:** smaller bottom wins (fewer, bigger pieces). $\frac{1}{3} > \frac{1}{5}$
- **Different tops and bottoms:** cross-multiply.

$\frac{3}{4}$? $\frac{5}{7}$ → $3 \times 7 = 21$, $5 \times 4 = 20$ → $\frac{3}{4} > \frac{5}{7}$

4. Adding and subtracting

Same bottom: add the tops, keep the bottom.

$$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$$



2 pieces + 3 pieces = 5 pieces, still sevenths.

Different bottoms: first make the bottoms the same (use the LCM), then add.

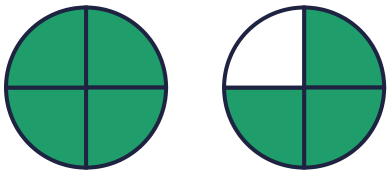
1. LCM of 2 and 3 is **6**
2. $\frac{1}{2} = \frac{3}{6}$, $\frac{1}{3} = \frac{2}{6}$
3. $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$



> **Common mistake** $\frac{1}{2} + \frac{1}{3}$ is **not** $\frac{2}{5}$. Never add the bottoms! Check: $\frac{1}{2}$ alone is already more than $\frac{2}{5}$.

5. More than one whole

When the top is bigger than the bottom, we have more than one whole:



$\frac{7}{4} = 1\frac{3}{4}$ (4 quarters make 1 whole, 3 quarters left)

Quick summary

Equal fractions	multiply or divide top and bottom by the same number
Simplest form	divide top and bottom by their HCF
Compare	same bottom → look at tops; otherwise cross-multiply
Add / subtract	make bottoms equal (LCM), then add / subtract the tops

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Fractions worksheet

Name: _____

Try every question. Show your working. Answers on the last page.

Date: _____

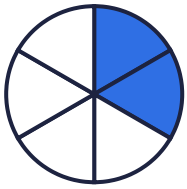
A. What fraction is shaded? Write it in simplest form.

1



Answer: _____

2



Answer: _____

3



Answer: _____

B. Fill in the box.

4 $\frac{2}{3} = \frac{\square}{12}$

5 $\frac{3}{5} = \frac{\square}{20}$

6 $\frac{7}{4} = \frac{\square}{16}$

C. Put <, > or = in the circle.

7 $\frac{3}{7} \bigcirc \frac{5}{7}$

8 $\frac{2}{3} \bigcirc \frac{3}{5}$

9 $\frac{4}{6} \bigcirc \frac{6}{9}$

D. Work out. Give the answer in simplest form.

10 $\frac{2}{9} + \frac{5}{9} =$ _____

11 $\frac{1}{4} + \frac{2}{3} =$ _____

12 $\frac{5}{6} - \frac{1}{4} =$ _____

E. Word problems.

13 Riya eats $\frac{3}{8}$ of a pizza and her brother eats $\frac{1}{4}$ of it. (a) How much of the pizza is eaten? (b) How much is left?

14 A ribbon is $\frac{3}{4}$ m long. Meena cuts off $\frac{1}{3}$ m. How long is the ribbon now?

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Answer key

With short working · every answer checked by code

A. Shaded

1. $\frac{3}{8}$

2. $\frac{2}{6} = \frac{1}{3}$

3. $\frac{5}{10} = \frac{1}{2}$

B. Equal fractions

4. $12 \div 3 = 4$, so $\frac{2}{3} = \frac{2\times4}{3\times4} = \frac{8}{12}$

5. $20 \div 5 = 4$, so $\frac{3}{5} = \frac{3\times4}{5\times4} = \frac{12}{20}$

6. $16 \div 4 = 4$, so $\frac{7}{4} = \frac{7\times4}{4\times4} = \frac{28}{16}$

C. Compare

7. same bottom, compare tops → $\frac{3}{7} < \frac{5}{7}$

8. cross-multiply: $2 \times 5 = 10$, $3 \times 3 = 9$ → $\frac{2}{3} > \frac{3}{5}$

9. cross-multiply: $4 \times 9 = 36$, $6 \times 6 = 36$ → $\frac{4}{6} = \frac{6}{9}$

D. Add / subtract

10. $\frac{2}{9} + \frac{5}{9} = \frac{7}{9}$

11. LCM = 12: $\frac{3}{12} + \frac{8}{12} = \frac{11}{12}$

12. LCM = 12: $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$

E. Word problems

13. (a) $\frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ eaten. (b) $1 - \frac{5}{8} = \frac{3}{8}$ left.

14. $\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$ m

Original questions by Adithya Vikram. Every answer is computed and checked by code (a second check with decimals) before it is printed.